

Perpendicularity of the Euler Line to the Segment Joining Feuerbach Points of a Triangle and Its Tangent-Reflection Triangle

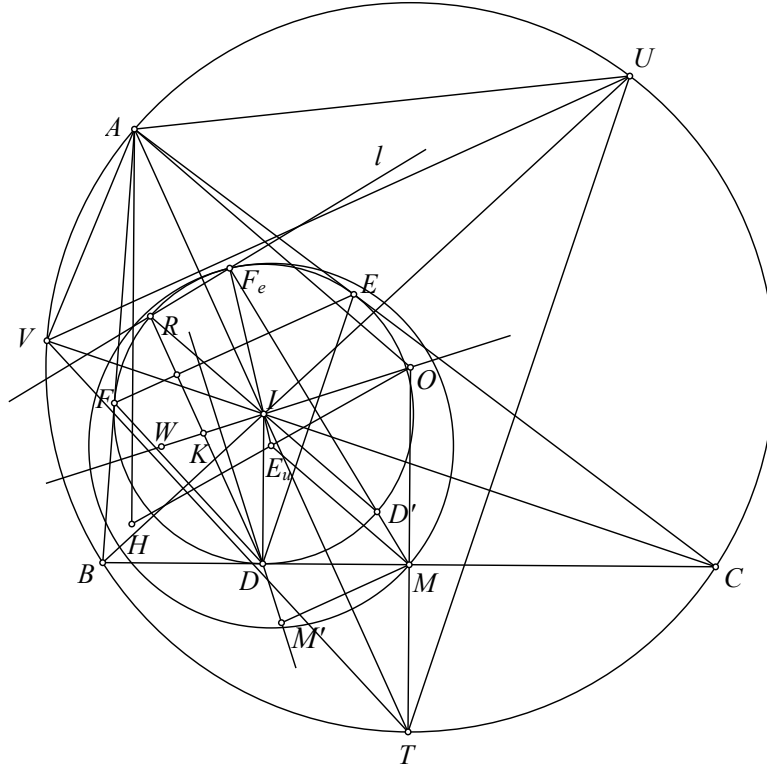
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Problem. Given a $\triangle ABC$ where ℓ is the tangent at the *Feuerbach* point F_e . The lines symmetric to ℓ with respect to BC , CA and AB form $\triangle XYZ$. Let F'_e be the *Feuerbach* point of $\triangle XYZ$. Prove that the Euler line of $\triangle ABC$ is perpendicular to the line $F_e F'_e$.

Solution. We first need several lemmas.

Lemma 1. Let $\triangle ABC$ have incircle (I) touching BC , CA , AB at D , E , F respectively. Let F_e be the *Feuerbach* point of $\triangle ABC$. Prove that F_e is the *Euler-Reflection* point of $\triangle DEF$.



Proof of Lemma 1. Let M be the midpoint of BC , let H and E_u be the orthocenter and nine-point center of $\triangle ABC$ respectively, let D' be the reflection of D over AI , and let AI meet the circumcircle (O) again at T . Since $AH \parallel ID$ and the pairs are isogonal with respect to $\angle BAC$ and $\angle DID'$, we have

$$\angle IAO = \angle HAI = \angle DIT = \angle TID'$$

so $ID' \parallel AO \parallel E_u M$. As F_e is the external center of similitude of (E_u) and (I) , the points M, D', F_e are collinear. Let M' be the reflection of M over AI . Then

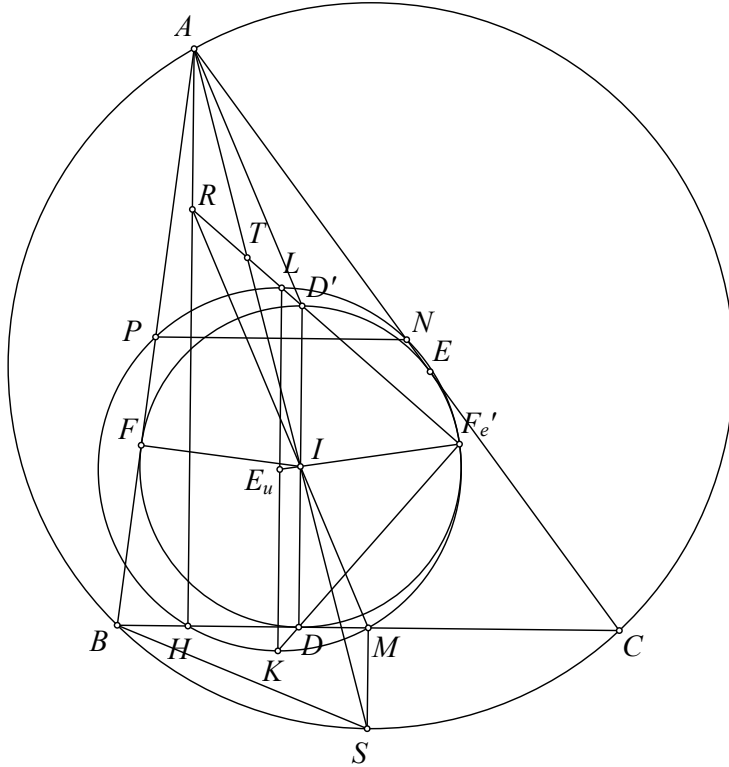
$$\frac{TI}{TO} = \frac{TB}{TO} = \frac{2TM}{TB} = \frac{MM'}{\sin \angle ATO \cdot TB} = \frac{MM'}{MD}$$

and since $\angle DMM' = 90^\circ - \angle M'MT = \angle MTI$, we have $\triangle IOT \sim \triangle M'DM$. From $MM' \perp IT$ and $DM \perp OT$, it follows that $DM' \perp OI$.

Let U, V be the midpoints of the minor arcs CA, AB respectively. Then $VA = VI$ and $UA = UI$, so UV is the perpendicular bisector of AI , hence $AI \perp UV$. Similarly, I is the orthocenter of $\triangle TUV$. Let K be the orthocenter of $\triangle DEF$. Since $EF \parallel UV \perp AI$, the triangles $\triangle DEF$ and $\triangle TUV$ have correspondingly parallel sides, so DT, EU, FV concur at W , the homothety center of the two triangles. Moreover, $\triangle DEF \cup \{I, K\} \sim \triangle TUV \cup \{O, I\}$, so W, K, I, O are collinear, i.e., IO is the *Euler* line of $\triangle ABC$.

Let R be the reflection of the orthocenter K of $\triangle DEF$ over EF , and let ℓ be the reflection of line IO over EF . Then R lies on both ℓ and ID' . Since $DM' \perp OI$, we have $MD' \perp \ell$. Combined with $\angle RF_e D' = 90^\circ$, it follows that F_e lies on ℓ . Therefore F_e is the *Euler-Reflection* point of $\triangle DEF$.

Lemma 2. Let $\triangle ABC$ be inscribed in (O) and circumscribed about (I) , touching BC, CA, AB at D, E, F respectively. Let M be the midpoint of BC , let MI meet the altitude from A at R , and let T be the midpoint of AI . Prove that R, T, F_e are collinear.



Proof of Lemma 2. Let N, P be the midpoints of AC, AB and let E_u be the circumcircle of $\triangle MNP$. Since (I) and (E_u) touch internally at F_e , F_e is the internal center of similitude. Draw diameter KL of (E_u) perpendicular to BC , so F_e, D, K are collinear and F_e, D', L are collinear with DD' a diameter of (I) .

The homothety at A with ratio 2 maps $N \mapsto C, P \mapsto B, T \mapsto I$, so T is the incenter of $\triangle ANP$. By *Protassov's* theorem applied to $\triangle ANP$, $F_e T$ bisects $\angle NF_e P$, hence $F_e T$ passes through L ,

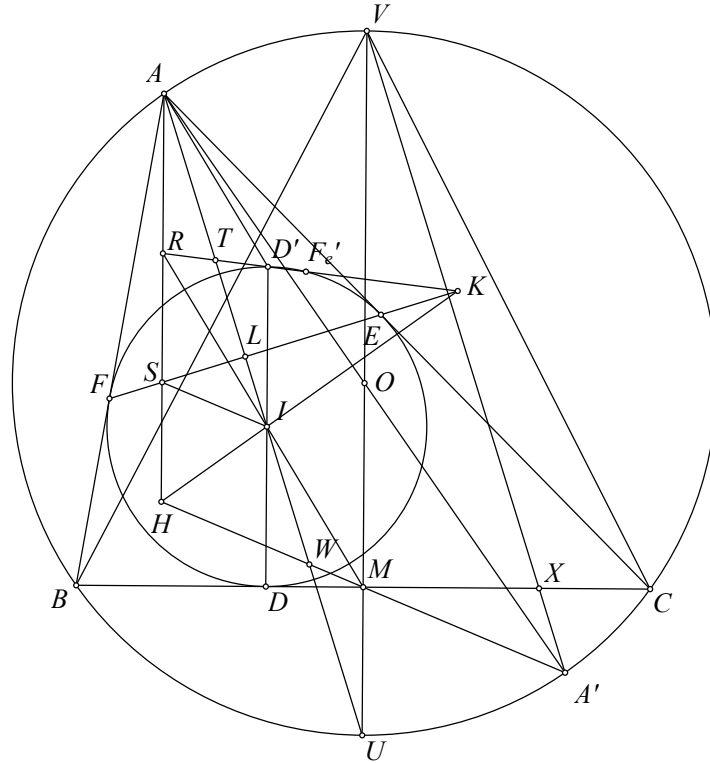
the midpoint of minor arc NP .

Let S be the midpoint of minor arc BC of (O) . Then $\triangle AIF \sim \triangle BSM$, so

$$\frac{AR}{MS} = \frac{AI}{IS} = \frac{IF}{MS} = \frac{ID'}{MS}$$

hence $AR = ID'$. Since $AR \parallel ID$, quadrilateral $IRAD'$ is a parallelogram. As T is the midpoint of AI , the points R, T, D' are collinear. Thus R, T, D', F_e are collinear.

Lemma 3. Let $\triangle ABC$ be inscribed in (O) with orthocenter H and circumscribed about (I) , touching BC, CA, AB at D, E, F respectively. Let F_e be the *Feuerbach* point of $\triangle ABC$ and let DD' be a diameter of (O) . Prove that $D'F_e, IH, EF$ are concurrent.



Proof of Lemma 3. Let T, L, M be the midpoints of AI, EF, BC respectively. Let IM meet AH at R . By Lemma 2, R, T, D', F_e are collinear. Let AH meet EF at S , and let U, V be the midpoints of the minor and major arcs BC of (O) . Draw diameter AA' of (O) , let $W = HM \cap AI$, and let $K = IH \cap EF$.

Since $AI \parallel A'V$ and $\triangle AEF \cup \{L, I\} \sim \triangle VCB \cup \{M, U\}$, we have

$$\frac{MW}{MH} = \frac{MW}{MA'} = \frac{MU}{MV} = \frac{LI}{LA} \quad (1).$$

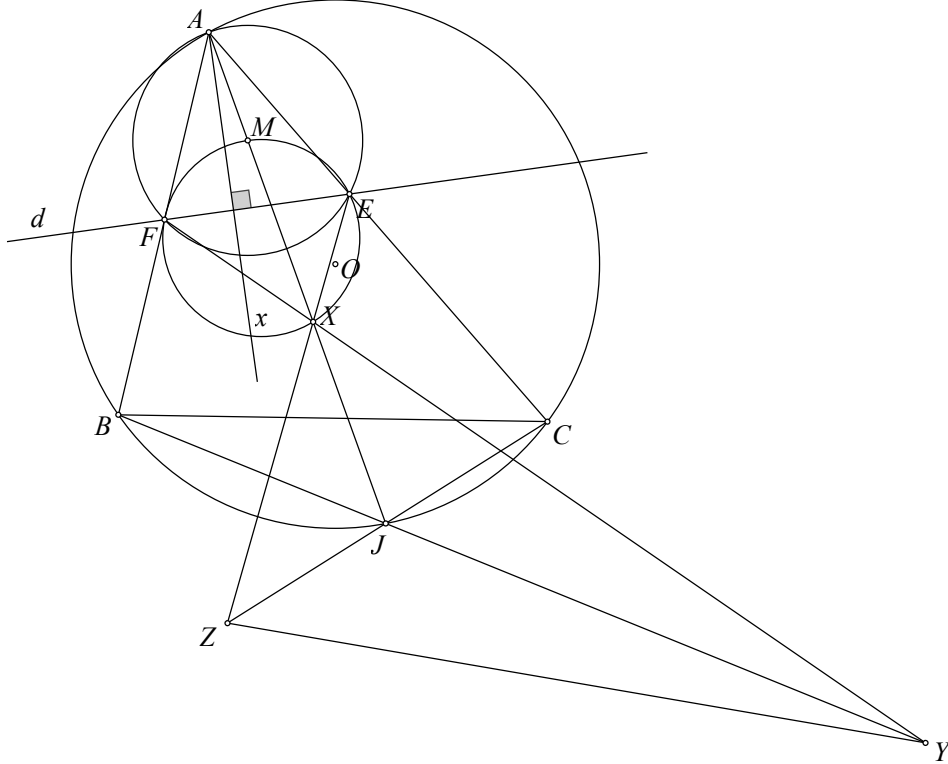
Let $X = VA' \cap BC$. Then $\angle BAS = \angle OAC = \angle A'VC$, so $\triangle AEF \cup \{I, S\} \sim \triangle VBC \cup \{X, U\}$. Hence $\angle AIS = \angle VA'H = \angle AWH$, so $SI \parallel HM$.

Applying *Menelaus'* theorem to $\triangle AHI$ with transversal S, L, K and to $\triangle AHW$ with transversal R, I, M gives

$$\frac{KI}{KH} = \frac{LI}{LA} \cdot \frac{SA}{SH} \stackrel{(1)}{=} \frac{MW}{MH} \cdot \frac{SA}{SH} \stackrel{SI \parallel HW}{=} \frac{MW}{MH} \cdot \frac{IA}{IW} = \frac{RA}{RH}.$$

By the converse of *Menelaus'* theorem on $\triangle AHI$, R, T, K are collinear. Therefore $D'F_e, IH, EF$ concur at K .

Lemma 4. Let $\triangle ABC$ be inscribed in (O) and let d be any line. The lines symmetric to d with respect to the sides of $\triangle ABC$ form $\triangle XYZ$. Prove that AX , BY , CZ concur at a point J on (O) , that J is the incenter of $\triangle XYZ$, and that the Steiner line of J with respect to $\triangle ABC$ is parallel to d .



Proof of Lemma 4. Let d meet AC and AB at E and F . By symmetry, FA and EA are the external bisectors of $\angle XFE$ and $\angle XEF$, so A is the excenter opposite X in $\triangle XEF$, hence XA bisects $\angle ZXY$. Similarly for the other sides. Thus AX , BY , CZ concur at the incenter J of $\triangle XYZ$.

Let M be the midpoint of arc EF of circle (XEF) , so M is the circumcenter of $\triangle AEF$. The ray from A perpendicular to EF is isogonal to AM in $\angle BAC$. The same holds for the isogonals of BJ and CJ with respect to the other angles. Therefore the Steiner line of J with respect to $\triangle ABC$ is parallel to d .

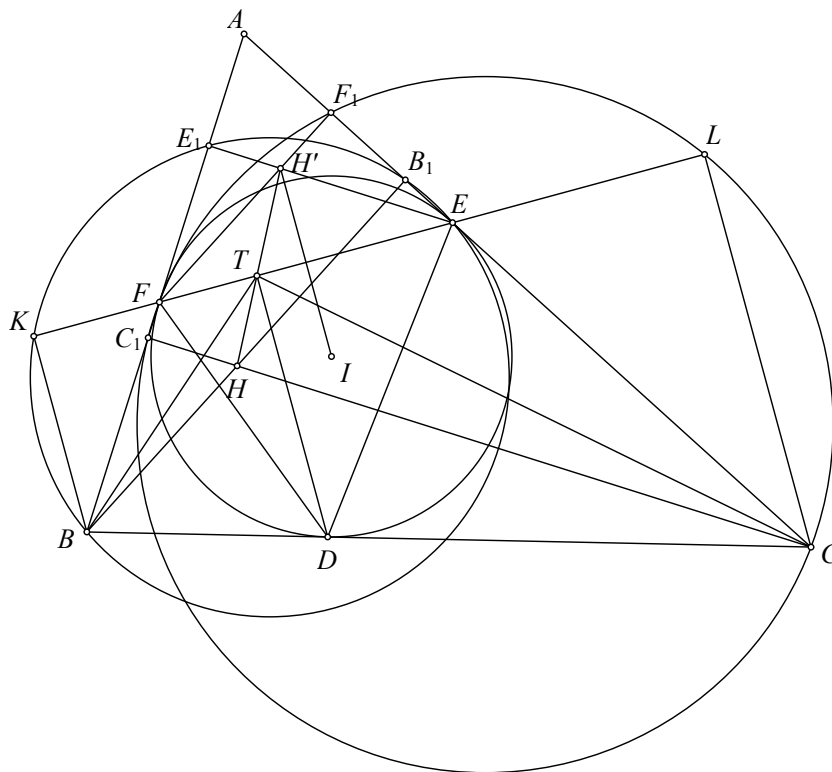
Lemma 5. Let $\triangle ABC$ have incircle (I) touching BC , CA , AB at D , E , F respectively. Let T be the foot of the perpendicular from D to EF , and let H' be the orthocenter of $\triangle AEF$. Prove that H , T , H' are collinear.

Proof of Lemma 5. Let CC_1 , BB_1 be the altitudes from C , B in $\triangle ABC$, and let EE_1 , FF_1 be the altitudes from E , F in $\triangle AEF$. The quadrilaterals C_1FF_1C and BE_1B_1E are cyclic with diameters CF and BE . Moreover,

$$\overline{H'E} \cdot \overline{H'E_1} = \overline{H'F} \cdot \overline{H'F_1} \quad \text{and} \quad \overline{HB} \cdot \overline{HB_1} = \overline{HC} \cdot \overline{HC_1},$$

so HH' is the radical axis of circles (BE) and (CF) . Let K , L be the feet from B , C to EF . Then K lies on (BE) and L lies on (CF) . Since $\angle KTB = \angle CTL$, we have $\triangle BKT \sim \triangle CLT$. As $DT \parallel BK \parallel CL \perp EF$,

$$\frac{TK}{TL} = \frac{KB}{LC} = \frac{FK}{EL} = \frac{TK - FK}{TL - EL} = \frac{TF}{TE}.$$

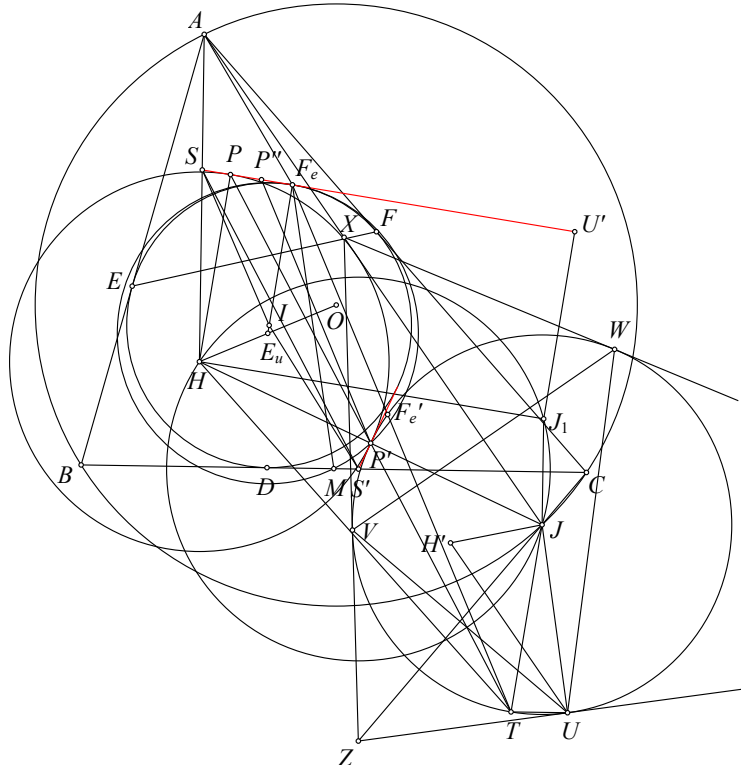


Back to the problem. We prove the theorem via the following observations.

We have $\triangle ATI \sim \triangle IDN$ where $N = AI \cap BC$. Since $GI^2 = GB^2 = GN \cdot GA$,

so T, D, G are collinear. Let DD' be a diameter of (I) . By *Lemma 3*, D', F_e, K are collinear, so $\angle DF_eK = 90^\circ = \angle DQK$ and quadrilateral DQF_eK is cyclic. By *Lemma 5*, $\frac{QF}{QE} = \frac{DB}{DC}$, so $\triangle I'FE \cup \{Q\} \sim \triangle GBC \cup \{D\}$. Hence

Observation 2. Let (J) be the incircle of $\triangle XYZ$. Then $HJ = 2R_{(J)} = 2d(H, \ell)$.



Proof of Observation 2. Redefine J as the reflection of H over P' , where P' is the reflection of P over IH , and let J_1 be the reflection of J over BC . Then S' is the circumcenter of $\triangle HJJ_1$, so

$$(J_1H, J_1J) = (S'H, S'P') = (SP, SH) = (\ell, AH).$$

Since $JJ_1 \parallel AH$, we have $HJ_1 \parallel \ell$. The same holds for reflections J_2, J_3 of J over CA, AB , so H, J_1, J_2, J_3 are collinear. By the converse of Steiner's theorem, J lies on (O) . By Lemma 4, J is the incenter of $\triangle XYZ$.

Let U, V, W be the points of tangency of (J) with YZ, ZX, XY , and let U' be the reflection of U over BC . Then YZ, ℓ, JU, J_1U' are symmetric with respect to BC , so $J_1U' \parallel HP \perp \ell$ and $HJ_1 \parallel \ell$. Thus quadrilateral $HPU'J_1$ is a parallelogram, whence $HP = J_1U' = JU = R_{(J)} = \frac{HJ}{2}$. It follows that P' lies on (J) .

Observation 3. Let T be the reflection of P over P' . Then $F_eT \perp OH$.

Proof of Observation 3. Let (E_u) be the nine-point circle of $\triangle ABC$. Since T is the reflection of P over P' and P' is the midpoint of HJ , $HPJT$ is a parallelogram, so $JT = HP = R_{(J)}$ and T lies on (J) . The line PF_e is a common external tangent to (E_u) and circle $(H; HP)$ at F_e and P . Let P'' be the midpoint of PF_e ; then P'' lies on their radical axis.

The homothety at H with ratio 2 maps $(E_u) \mapsto (O)$ and $P' \mapsto J$, so P' lies on both (E_u) and $(H; HP)$. Hence $P'P''$ is the radical axis of (E_u) and $(H; HP)$, so $P'P'' \perp OH$. As $P'P''$ is the midline of $\triangle PF_eT$, we obtain $F_eT \perp OH$.

Observation 4. $F_e'T \perp OH$.

Proof of Observation 4. By Lemma 4 and angle chasing, $\angle VUW = \angle ABC$, so $\triangle UVW \cup \{H'\} \sim \triangle ABC \cup \{H\}$, where H' is the orthocenter of $\triangle UVW$. Moreover,

$$\overrightarrow{TU} = \overrightarrow{TJ} + \overrightarrow{JU} = \overrightarrow{J_1U'} + \overrightarrow{JU},$$

and since J_1U' and JU are symmetric with respect to BC , we have $TU \parallel BC$. Thus

$$(F_e'T, BC) = \angle F_e'TU = \angle UHJ.$$

By Lemma 1, $H'J$ and $H'U$ are the Steiner lines of F_e' and U with respect to $\triangle UVW$. Since $\angle UH'J = \angle AHO = 90^\circ + (OH, BC)$, it follows that $OH \perp F_e'T$. Therefore the points F_e, F_e', T lie on a line perpendicular to OH .